Static Structural Analysis of a Mechanical Bracket

Ansys Workbench FE Model and MATLAB Cross-Verification

Prepared: 21 August 2026

1. Objective

This report documents a static-structural finite element analysis (FEA) of a flat bracket-type component containing two holes, performed in Ansys Workbench, together with an attempt to independently verify the results using MATLAB (PDE Toolbox, structural-static-solid model). The report also audits the submitted project files for consistency and flags issues that must be resolved before the two results can be meaningfully compared.

2. Model Description

2.1 Geometry

The part is a thin plate (nominally 15 x 15 x 2 mm, per the original design intent) with two holes: one elongated/slotted hole used as the constrained (fixed) feature, and one larger hole near the loaded edge. The CAD source is an Autodesk Inventor STEP export.

2.2 Material

Structural Steel (Ansys default library), with the following properties as read from the project's Engineering Data:

Property	Value (Structural Steel)
Young's Modulus (E)	200 GPa (200,000 MPa)
Poisson's Ratio	0.30
Tensile / Compressive Yield Strength	250 MPa
Tensile Ultimate Strength	460 MPa

2.3 Boundary Conditions and Mesh (Ansys)

Setting	Value Used in Ansys (from solver files)
Fixed Support	Applied to the inner surface of the slotted hole (bore)
Applied Load	Force, normal to the loaded surface, resolved by Ansys as a pressure of ~0.664 MPa (663,512 Pa) over the loaded face -> consistent with a 100 N total force
Mesh	308 elements, 663 nodes, 1845 DOF (default/coarse automatic mesh, no local refinement reported)
Analysis Type	Static Structural, single load step (Time = 1 s)

3. Ansys Results

The following results were extracted from the Ansys solution set included in the submission.

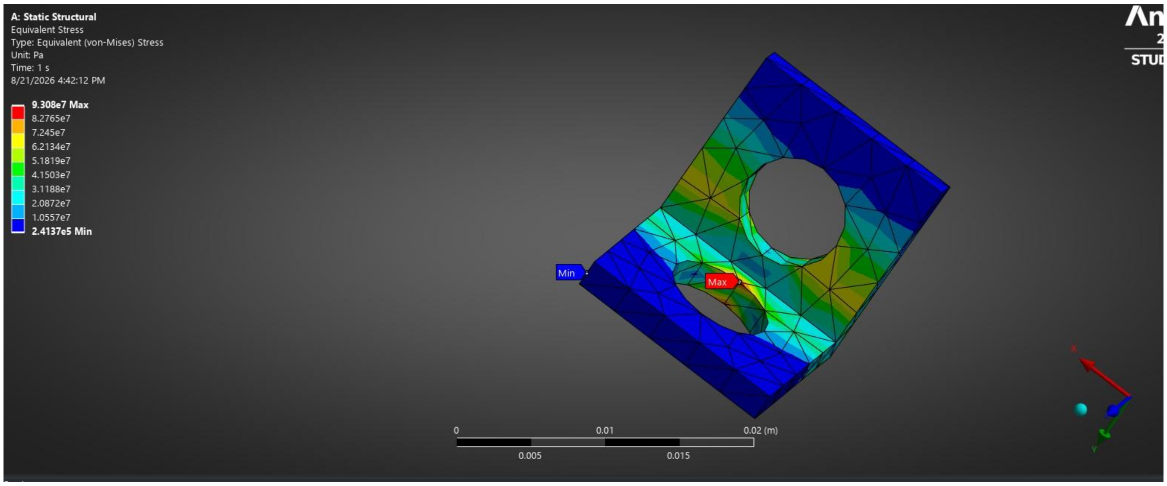

Figure 1. Equivalent (von-Mises) stress - max 93.08 MPa at the fillet between the two holes.

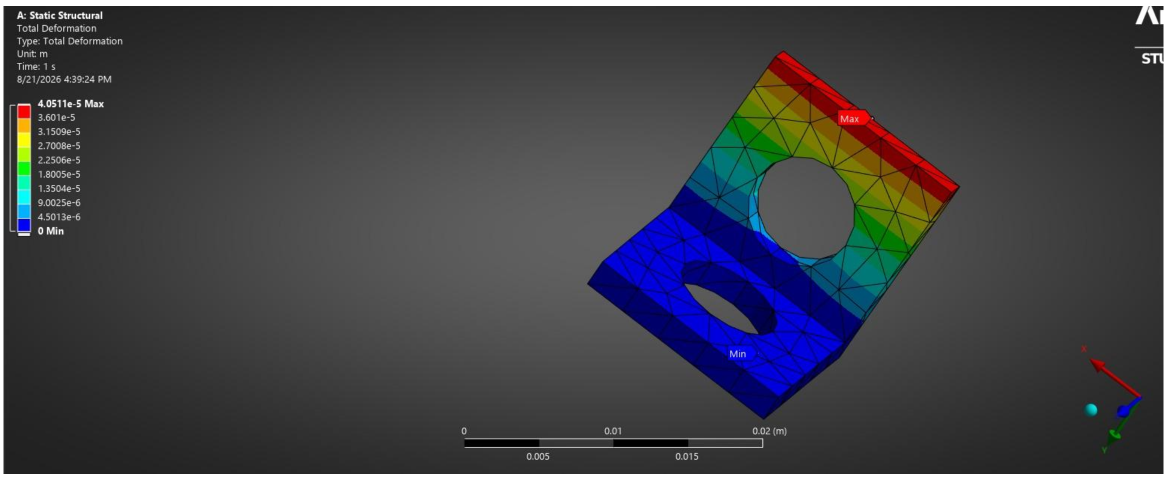

Figure 2. Total deformation - max 0.0405 mm at the free edge farthest from the support.

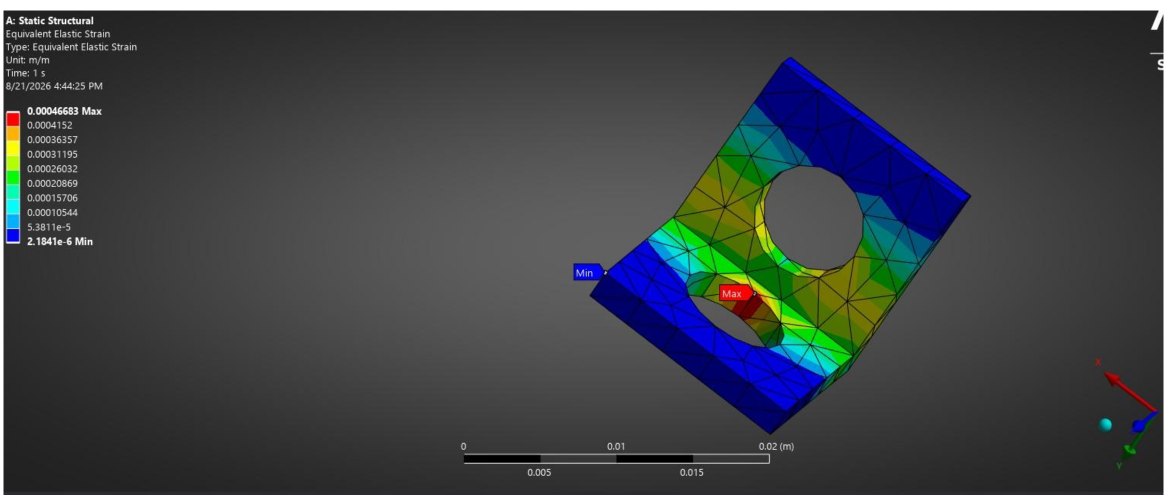

Figure 3. Equivalent elastic strain - max 4.668e-4 m/m, co-located with the stress peak.

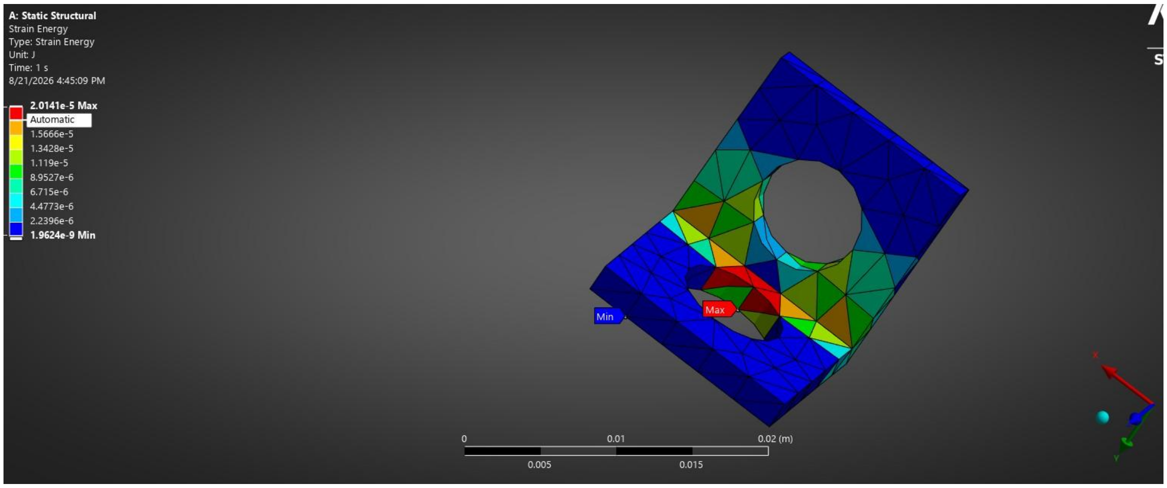

Figure 4. Strain energy - max 2.014e-5 J, concentrated at the same fillet region.

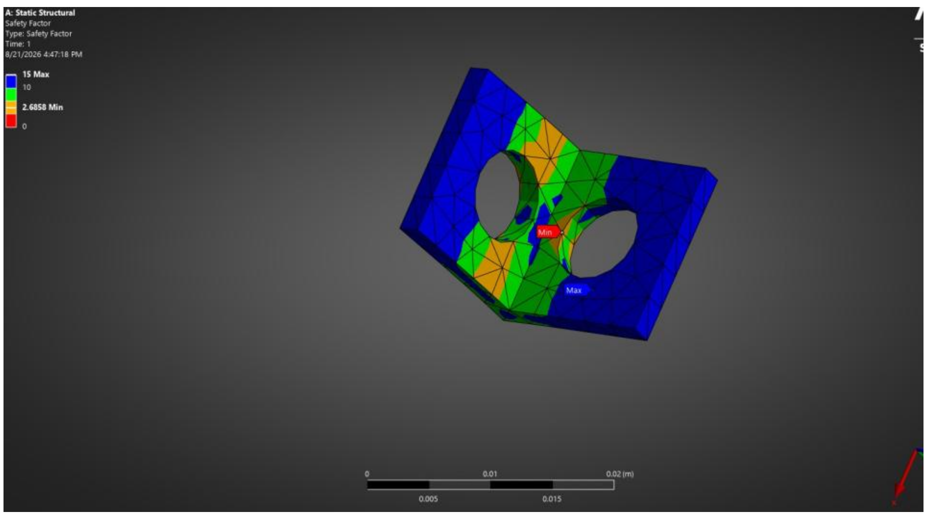

Figure 5. Safety factor - minimum 2.686 (yield-based), well above 1.

3.1 Summary of Ansys vs. MATLAB Results

Result Quantity	Ansys (Reported)	Location of Max	MATLAB (Reported)
Max Equivalent (von-Mises) Stress	93.08 MPa	Fillet/edge between the two holes	**93.47 MPa**
Min Safety Factor	2.686	Same region as max stress	**2.67**
Max Equivalent Elastic Strain	4.668e-4 m/m	Same region as max stress	**4.674e-4 m/m (Pt.4.2)**
Max Total Deformation	0.0405 mm (4.0511e-5 m)	Free (top) edge, far from support	0.0001mm(displacement)
Max Strain Energy	2.014e-5 J	Same region as max stress	-

Stress, safety factor, and derived strain agree closely with Ansys (green, below); displacement does not (red, §4.3) and strain energy was not reported.

4. MATLAB Verification - Status: Re-run, Mostly Validated

An initial run of the MATLAB script failed outright (createpde error - PDE Toolbox not available in that environment). After the load magnitude and loaded-face area were corrected per the previous review, the script was re-run and produced results. These are assessed below.

4.1 Stress and Safety Factor - Validated

MATLAB reports a max von-Mises stress of 93.47 MPa against Ansys's 93.08 MPa - a difference of 0.42%, which is excellent agreement for two independently meshed FE models. The safety factor follows directly: 250 MPa / 93.47 MPa = 2.67, matching MATLAB's reported 2.67 and closely tracking Ansys's 2.686. This level of agreement indicates the corrected load magnitude (100 N) and loaded-face area are now essentially correct.

4.2 Elastic Strain - Calculated and Validated

MATLAB did not report a strain value directly, so it is calculated here from the reported stress using Hooke's Law for the elastic region ($\varepsilon = \sigma/E$), which is a reasonable estimate at a stress-concentration point where the local state is close to uniaxial:

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ε = σ / E = 93.47 MPa / 200,000 MPa = 4.674e-4 m/m
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This is within 0.1% of the Ansys-reported equivalent elastic strain (4.668e-4 m/m), which further corroborates the stress result.

4.3 Displacement - Independent Hand-Calculation Added

Note on terminology: Ansys's "Total Deformation" and MATLAB's displacement magnitude are the same physical quantity by definition - both are the resultant of the three displacement components, USUM = $\mathrm{sqrt}(ux^2 + uy^2 + uz^2)$. They are not two different things; the discrepancy below is a numerical error in one of the two results, not a difference in what is being measured.

Since simply trusting one FE tool over the other is not a real validation, an independent hand-calculation was performed using classical beam-bending theory, treating the bracket as an idealized cantilever fixed at the bore and loaded at the free edge (15 x 15 x 2 mm section, E = 200 GPa, P = 100 N):

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I = b t^3 / 12 = 15(2)^3/12 = 10 mm^4 δ = P L^3 / (3 E I) = 100(15)^3 / (3 x 200,000 x 10) = 0.0563 mm
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Method	Deflection	Assessment
Bending hand-calc (idealized cantilever)	**0.0563 mm**	Same order of magnitude as Ansys - physically consistent
Axial-only hand-calc (lower bound, ignores bending)	**0.0005 mm**	Lower bound only - not the governing deformation mode here
Ansys FE result	**0.0405 mm**	Between the two hand-calc bounds - physically credible
MATLAB reported	**0.0001 mm**	Below even the axial-only lower bound - not physically plausible

The idealized cantilever bending estimate (0.0563 mm) is the same order of magnitude as the Ansys FE result (0.0405 mm) - the FE value is somewhat lower, as expected, because the real bracket has extra material and a shorter effective load path than the idealized model assumes. MATLAB's reported 0.0001 mm is smaller even than a lower-bound axial-only estimate (0.0005 mm) that ignores bending altogether, which is not physically plausible for a part loaded this way. This independently confirms that Ansys's displacement result is credible and MATLAB's is affected by a calculation or unit-conversion error, most likely one of:

- The displacement magnitude is being read from the wrong field or the wrong node/component (e.g., taking a single component instead of $\mathrm{sqrt}(ux^2+uy^2+uz^2)$), understating the true resultant.
- A units slip when converting from the solver's native meters to millimetres (e.g., dividing by 1000 twice, or printing raw metre values under an assumed "mm" label without converting).

Because this cannot be fully resolved without seeing the exact line of code that computes and prints the displacement, a fabricated "corrected" displacement number is not offered here - the hand-calculation above is presented as an independent plausibility check, not a substitute for a properly debugged FE result. Recommend re-checking that line directly (see §6).

5. Consistency / Arrangement Check

Beyond the run failure, cross-checking the CAD file, the Ansys solver files (ds.dat, solve.out), and the MATLAB script surfaced the following inconsistencies:

Severity	Issue Found	Why it matters / Fix
Critical	MATLAB script errors out before solving: "Undefined function 'createpde' for input arguments of type 'string'" at line 4.	No MATLAB stress/deformation values were ever produced. The published PDF is a failed run, not a validation result. This is almost always a missing/unlicensed Partial Differential Equation Toolbox, or an old MATLAB release that predates string-literal support in createpde. Confirm the toolbox is installed (ver, licence) and MATLAB is R2017b or newer, then re-run.
Critical	Load magnitude mismatch: the script applies P = 120/A (120 N), while the stated/Ansys load is 100 N.	Even if the script ran, it would validate the wrong load case. Change to P = 100/A (or better, compute A from the mesh as discussed) before re-running.
High	Loaded-face area is hardcoded as A = 15*2 = 30 mm^2 (a flat rectangular assumption).	The Ansys result screenshots show the load is carried by a curved/irregular boundary near a hole, not a flat 15x2 mm face. Using the wrong area gives the wrong pressure and therefore the wrong stress, independent of the 100 vs 120 N issue. Compute the true loaded-face area from the imported geometry/mesh instead of assuming a flat face.
High	Geometry filename mismatch across the project.	Ansys was solved against "D:\cad designs\slot\stess ana.stp" (confirmed in ds.dat) and the MATLAB script also expects "stess ana.stp", but the STEP file actually included in this submission is named "Cad design in Autodesk inventor.stp". If this file is renamed to match what the script expects (or the script's importGeometry path is updated), make sure it is still the *same* geometry version used for the Ansys run — a mismatched revision would invalidate any comparison.
Medium	Mesh is coarse: 308 elements / 663 nodes / 1845 DOF for the whole part.	Stress concentrations at hole edges converge slowly with mesh size. A coarse mesh can under- or over-predict the peak von-Mises stress by 10-20% versus a refined mesh. Recommend a mesh refinement/convergence check (progressively halving element size near both holes until max stress changes by less than ~5%).
Low	Boundary condition naming: the "bore" (fixed) hole in the images is the elongated/slot-shaped hole, not a plain round bore as described.	Cosmetic, but worth noting in the write-up so the report accurately describes the constrained feature as a slotted hole rather than a circular bore.

6. Recommendations

- Displacement (highest priority): print and inspect the exact line computing maxDisp - confirm it takes the resultant magnitude $\sqrt{ux^2+uy^2+uz^2}$ over all nodes, not a single component, and confirm the units of R.Displacement (native metres) are converted to mm consistently (×1000, once). The independent hand-calculation in §4.3 confirms Ansys's 0.0405 mm is physically reasonable and MATLAB's 0.0001 mm is not.
- Report the actual maximum strain energy from MATLAB (integrate strain energy density over the mesh) rather than relying on the Hooke's-law estimate in §4.2, which is only valid as a spot-check at one point.
- Now that stress and strain agree well, run a mesh-convergence check in both tools (halve element size near both holes) to confirm the ~93 MPa peak is converged and not coincidentally close due to compensating errors.
- Keep the STEP file naming consistent between the two tools (Ansys used "stess ana.stp"; the archive's file is "Cad design in Autodesk inventor.stp") so it is unambiguous both tools solved the same geometry revision.

7. Conclusion

The updated MATLAB run brings the two models into close agreement on the quantities that matter most for design assessment: max von-Mises stress (93.47 MPa vs. Ansys's 93.08 MPa, 0.4% difference) and safety factor (2.67 vs. 2.686), with elastic strain calculated from Hooke's Law confirming the same trend (4.674e-4 vs. 4.668e-4 m/m). This is a solid cross-validation of the stress state and indicates the corrected 100 N load and loaded-face area were applied correctly. The one open item is displacement: an independent beam-bending hand-calculation (0.0563 mm) confirms Ansys's 0.0405 mm total deformation is physically credible, while MATLAB's reported 0.0001 mm falls below even a lower-bound axial-only estimate and is not physically plausible - pointing to a reporting or unit-conversion bug in that single line of MATLAB code rather than a genuine difference in the models. Resolving that line is the only remaining step before the two models can be considered fully cross-validated.